

The Cauchy-Schwartz Inequalities

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In real analysis we frequently use inequalities coming from geometry. The **Cauchy-Schwartz inequality** states that for two vectors $\mathbf{a} = \sum_{i=1}^n a_i \mathbf{e}_i$ and $\mathbf{b} = \sum_{i=1}^n b_i \mathbf{e}_i$ equipped with the euclidean dot product (so that $\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v}$) we have

$$\mathbf{a}^2 \mathbf{b}^2 \geq (\mathbf{a} \cdot \mathbf{b})^2$$

with equality achieved precisely when $\mathbf{a}$ is parallel to $\mathbf{b}$. We can prove this identity for all n simultaneously by considering the quadratic function

$$y(x) = (a_1^2 + a_2^2 + \dots + a_n^2)x^2 + 2(a_1b_1 + a_2b_2 + \dots + a_nb_n)x + b_1^2 + b_2^2 + \dots + b_n^2$$

which can be rewritten as

$$y(x) = (a_1x + b_1)^2 + (a_2x + b_2)^2 + \dots + (a_nx + b_n)^2.$$

When our coefficients come from $\mathbb{R}$ then we have a sum of nonnegative real numbers, so this expression can have at most one root. We can restate this fact using the discriminant

$$4(a_1b_1 + a_2b_2 + \dots + a_nb_n) - 4(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \leq 0$$

which turns into the Cauchy-Schwartz inequality after some rearrangement.

To show that we achieve equality if and only if $\mathbf{a}$ is parallel to $\mathbf{b}$, we notice that equality is achieved if $y(x) = 0$ is possible over the real numbers. That would imply the system of equations $a_1x + b_1 = 0$, $a_2x + b_2 = 0$, ..., $a_nx + b_n = 0$ has a unique solution, and therefore $-b_i/a_i$ is constant for $i = 1, 2, \dots, n$.

Exercise 1 Use the Cauchy-Schwartz inequality to prove the triangle inequality

$$\sqrt{(\mathbf{a} + \mathbf{b})^2} \leq \sqrt{\mathbf{a}^2} + \sqrt{\mathbf{b}^2}.$$

This is used to conclude that Euclidean distance is well defined.

Exercise 2 For another proof of the Cauchy Schwartz inequality limited to two dimensions, using the complex numbers $z_1 = x_1 + y_1i$ and $z_2 = x_2 + y_2i$, expand the product $z_1 \bar{z}_2 \bar{z}_1 z_2$ in two different ways.

Exercise 3 For another proof of the Cauchy Schwartz inequality limited to three dimensions, using the standard Euclidean cross product $\mathbf{a} \times \mathbf{b} = (a_2b_3 + a_3b_2)\mathbf{e}_1 - (a_1b_3 + a_3b_1)\mathbf{e}_2 + (a_1b_2 - a_2b_1)\mathbf{e}_3$, show that

$$\mathbf{a}^2 \mathbf{b}^2 = (\mathbf{a} \cdot \mathbf{b})^2 + (\mathbf{a} \times \mathbf{b})^2.$$

That is far from the end of the story. (Fortunately) we do not live in a Euclidean reality, so we need to understand the geometry of non euclidean spaces. We will define $(n-1, 1)$ -space, or n dimensional minkowski space to be an n dimensional vector space equipped with the

dot product $\mathbf{a} \circ \mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_{n-1}b_{n-1} - a_nb_n$. Notice that the quadratic form induced by this dot product is not positive definite. We will still write $\mathbf{v} \circ \mathbf{v} = \mathbf{v}^2$ but with the understanding that we are squaring in Minkowski space until stated otherwise.

The classical Caley-Klein model of the hyperbolic plane is isomorphic to projective (2,1) space. The equation $x^2 + y^2 - z^2 = -1$ is called a pseudosphere because it is seen as a sphere of imaginary radius, and we typically imagine hyperbolic geometry happening on its surface. More people are probably familiar with the poincare disk model of the hyperbolic plane or the upper half complex plane models of the pseudosphere, because they are conformal in a way that is intuitive to us. In any case we will want to prove a variant of the Cauchy-Schwartz inequality if we wanted to measure distances on the upper hemisphere of the pseudo sphere.

The **hyperbolic Cauchy-Schwartz inequality** states that whenever $\mathbf{a}^2, \mathbf{b}^2 < 0$ we have

$$\mathbf{a}^2 \mathbf{b}^2 \leq (\mathbf{a} \circ \mathbf{b})^2$$

which can be proved in a very similar manner to the Euclidean version. This time we consider the quadratic

$$y(x) = (a_1^2 + a_2^2 + \dots + a_{n-1}^2 - a_n^2)x^2 + 2(a_1b_1 + a_2b_2 + \dots + a_{n-1}b_{n-1} - a_nb_n)x + b_1^2 + b_2^2 + \dots + b_{n-1}^2 - b_n^2.$$

We can factor this into $y = (a_1x + b_1)^2 + (a_2x + b_2)^2 + \dots + (a_{n-1}x + b_{n-1})^2 - (a_nx + b_n)^2$ by completing the square. Since $\mathbf{a}^2 < 0$ we know this parabola opens downward, and by setting $x = -b_n/a_n$ we get $y(-b_n/a_n) \geq 0$. Therefore this quadratic has at least one root, and we can write out an inequality involving the discriminant

$$4(a_1b_1 + a_2b_2 + \dots + a_{n-1}b_{n-1} - a_nb_n) - 4(a_1^2 + a_2^2 + \dots + a_{n-1}^2 - a_n^2)(b_1^2 + b_2^2 + \dots + b_{n-1}^2 - b_n^2) \geq 0$$

which turns into the hyperbolic Cauchy-Schwartz inequality after some rearrangement. The argument that equality occurs when the vectors are parallel is similar to the previous instance.

Exercise 4 Use the hyperbolic Cauchy-Schwartz inequality to prove the hyperbolic triangle inequality

$$\sqrt{|(\mathbf{a} + \mathbf{b})^2|} \geq \sqrt{|\mathbf{a}^2|} + \sqrt{|\mathbf{b}^2|}$$

whenever $\mathbf{a}^2, \mathbf{b}^2 < 0$.

Exercise 5 For another proof of the hyperbolic Cauchy-Schwartz inequality in (1,1)-space, consider expressions of the form $z_1 = x_1 + y_1j$, $z_2 = x_2 + y_2j$ in $\mathbb{R}[j]/(j^2 - 1)$ (the commutative ring extension obtained from adjoining j to $\mathbb{R}$ subject to the constraint $j \neq \pm 1$, and $j^2 = 1$). We get the inequality by expanding the product $z_1\bar{z}_2\bar{z}_1z_2$ in two different ways.

Exercise 6 For another proof of the hyperbolic Cauchy-Schwartz identity in (2,1)-space we define the **hyperbolic cross product** as $\mathbf{a} \times \mathbf{b} = (a_2b_3 + a_3b_2)\mathbf{e}_1 - (a_1b_3 - a_3b_1)\mathbf{e}_2 - (a_1b_2 - a_2b_1)\mathbf{e}_3$. The reader can check that $\mathbf{a} \circ (\mathbf{a} \otimes \mathbf{b}) = \mathbf{b} \circ (\mathbf{a} \otimes \mathbf{b}) = 0$ and as well as

$$\mathbf{a}^2 \mathbf{b}^2 = (\mathbf{a} \circ \mathbf{b})^2 - (\mathbf{a} \otimes \mathbf{b})^2.$$